the brakes. Hence the efficiency of the brakes decreases or becomes inactive. Therefore, at the particular interval of time the brake system has to be checked and adjusted for correct action if necessary. The adjustments in the brake system can be done by the following two modes:

1. Minor adjustment
2. Major adjustment

6.9.1 Minor Adjustment

In this method, without removing the wheels, the adjustments are made in the brake shoes upto the extent till the wear occurs. Generally in the vehicles front wheel brake shoes wear faster than the brake shoes in the rear wheels. Hence the brake drum has to be removed from the wheels and checked. If there are no problems (as stated below) then a small adjustment is enough.

i) Wear occurred in brake drum or brake drum leaves the center position.
ii) Brake shoes are soaked in brake fluid
iii) The brake shoes get wearer and the top of the rivet is visible outside the brake shoes.

6.9.2 Major Adjustment

In this method all the brake drum and brake shoes must be checked by removing all the brake drums. If there is any wear found in the brake drum or if the brake drum is shifted from the center position, then it has to be corrected by using the brake drum lathe and used. If the wear of the brake drum is more, then the drum must be replaced by a new drum. If there is any wear occurred in brake shoes for about 1 mm or above that, then the brake shoe must be replaced by a new one. If it is a hydraulic braking system then the paths (through which the brake fluid is circulated) have to be cleaned. Used parts of the wheel cylinder and master cylinder must be replaced by new parts. Required amount of brake fluid must be refilled.

Anchor pin and brake shoes must be corrected according to the requirement. The movement of the brake pedal should also be checked and adjusted as per the requirement. For more detailed procedures on working methods, the manual supplied by the vehicle manufacturers must be referred.

6.10 Brake pedal free play

When the brake pedal is pressed, the parts of the brake will not respond to function. First, the brake pedal will move to a certain distance. This distance is named as brake pedal free play. After this movement, further pressing of the brake pedal starts the parts of the brakes to function and stops the wheel. The "motionless movement" of the brake pedal is generally less than 10 mm. Driving and stopping the vehicles with control can be possible only by understanding this "motionless movement".

6.11 Brake Efficiency

To stop the vehicle the maximum braking force or retarding force (F) is applied on the wheel. This braking force depends on the μ- called as the co efficient of friction between the tyre tread and the road and the total load (W) acting on the vehicle. That is

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